EEG-to-Image Retrieval via Asymmetric Brain–Image Co-Adaptation

Chi Kit Wong
Sida Zhang
Zhiyi Chen
Ziyi Guo
AI Thrust
HKUST (GZ)
{ckwong627, szhang644, zchen986, zguo315}@connect.hkust-gz.edu.cn

Project repository: Chikit-WONG/Human-Centered_AI_Group_Project

Abstract

Electroencephalography (EEG) offers a non-invasive view of millisecond-scale neural responses, but exact-image retrieval remains difficult because EEG is noisy, subject-dependent, and semantically misaligned with pretrained image features. We study 200-way zero-shot EEG-to-image retrieval on THINGS-EEG2 and propose an asymmetric co-adaptation strategy: a lightweight EEG encoder is trained rapidly while low-rank adapters update a pretrained CLIP image encoder at a slower rate. Across ten subjects and five fixed seeds, our method obtains $86.66 \pm 0.69\%$ Top-1 and $98.38 \pm 0.14\%$ Top-5 accuracy. A strictly paired subject-08 ablation improves Top-1 from 29.5% with a frozen image encoder to 61.5% with equal-rate adaptation and 89.0% with asymmetric adaptation, isolating the contribution of co-adaptation from the EEG architecture. A focused hyperparameter study raises the seed-42 ten-subject Top-1 mean from 85.85% to 87.70%, while strict leave-one-subject-out evaluation reaches 20.40% Top-1 and exposes a substantial subject-shift gap. Finally, controlled matching experiments distinguish standard independent retrieval from batch-level assignment. Hungarian and Sinkhorn decoding can improve a known one-to-one session, but their advantage weakens when images are missing or multiple real EEG measurements correspond to one image. These results support asymmetric representation learning while clarifying the assumptions and limitations of session-aware post-processing.

1 Introduction

Human visual experience produces rapid neural activity that can be measured non-invasively with EEG. The resulting signals have high temporal resolution and are comparatively accessible for human–computer interaction, but their low signal-to-noise ratio, limited paired data, and strong inter-person variability make visual decoding difficult. THINGS-EEG2 provides repeated EEG responses to natural object images and has enabled controlled evaluation of neural representations at scale [1].

We focus on *exact EEG-to-image retrieval*: given the EEG response evoked by an unseen stimulus, rank the correct image among 200 candidates. This differs from image generation and from category-level recognition. The output is a ranked gallery, and the primary metrics are independent-query Top-1 and Top-5 accuracy. A central challenge is the representation mismatch between a small, noisy EEG modality and an image encoder pretrained on large visual corpora. Training only

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an EEG encoder forces neural signals into a fixed machine-vision space; fully fine-tuning the image encoder risks destroying useful pretrained structure.

Our method addresses this tension with asymmetric co-adaptation. The EEG encoder and image-side LoRA adapters [2] share the same contrastive objective, but the EEG branch receives a larger learning rate. This design is motivated by two-time-scale optimization [3]: the randomly initialized EEG branch must learn rapidly, whereas the pretrained image branch should move conservatively. Figure 1 shows both training and retrieval. We make four contributions:

- an efficient dual-side alignment framework that combines a compact EEG encoder with conservative CLIP image-side adaptation;
- a strictly paired A/B/C ablation that holds the EEG architecture, initialization, data, seed, and training budget fixed while changing only image-side optimization;
- subject-dependent five-seed evaluation, a focused rank–learning-rate study, and strict leave-one-subject-out (LOSO) evaluation;
- a fairness-controlled comparison of five decoders on identical score matrices, including robustness tests that deliberately violate one-to-one correspondence.

The distinction between representation quality and assignment post-processing is important. Independent retrieval asks which image best matches an EEG query. In contrast, Hungarian, stable, and transport-based decoders use information about the whole evaluation batch. We therefore report these quantities separately and do not relabel assignment accuracy as standard retrieval Top-1 or Top-5.

2 Related Work

EEG–image alignment. NICE demonstrated that contrastive alignment can learn image-related EEG representations for zero-shot object recognition [4]. ATM-S introduced temporal and spatial attention and used EEG embeddings for retrieval and guided image reconstruction [5]. UBP reduces the vision–brain information gap by adapting image content with an uncertainty-aware blur prior [6]. NeuroBridge uses cognitive-prior augmentations and a shared semantic projector for bidirectional alignment [7]. Shallow Alignment argues that intermediate image features better match the granularity of neural signals than final semantic features [8]. These approaches motivate a broader question: should the image target remain fixed, or should it adapt jointly with the brain representation? Our work tests the latter under a deliberately conservative update rule.

Pretrained image representations and parameter-efficient adaptation. CLIP learns transferable image–text representations from large-scale supervision [9]. We use its ViT-B/32 image encoder as the visual prior. LoRA freezes the original weight matrix and optimizes a low-rank residual, reducing trainable parameters and limiting drift [2]. Our use of LoRA is modality-specific: the adapters reshape the image target only enough to improve EEG alignment, while the EEG branch remains the primary learner.

Global assignment and optimal transport. The Hungarian algorithm solves a linear one-to-one assignment problem [10]; Gale–Shapley stable matching produces a stable pairing from ranked preferences [11]; and Sinkhorn iterations approximate entropy-regularized optimal transport by matrix scaling [12, 13]. Such decoders can exploit known batch structure, but they change the inference problem. We evaluate them as optional offline analyses, not as replacements for independent retrieval.

3 Task and Method

3.1 Retrieval task

Let $x_i \in \mathbb{R}^{17\times 250}$ denote the preprocessed EEG response for query i, and let v_j denote candidate image j. The EEG encoder f_θ and image encoder g_ϕ produce L2-normalized embeddings

$$z_i = \frac{f_\theta(x_i)}{\|f_\theta(x_i)\|_2}, \qquad y_j = \frac{g_\phi(v_j)}{\|g_\phi(v_j)\|_2}. \tag{1}$$

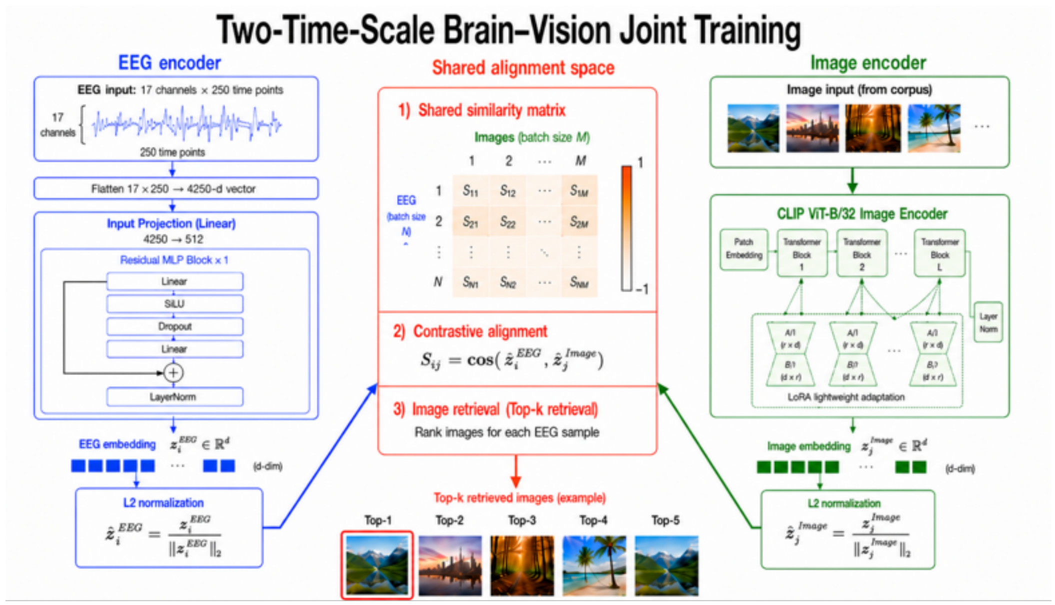

Figure 1: Detailed architecture of our method. During training, a compact EEG encoder and image-side LoRA adapters map paired EEG trials and stimulus images into a shared CLIP-aligned space and are optimized with a contrastive loss. The EEG branch learns at a faster rate than the image branch. During inference, a query EEG embedding is compared with a precomputed image bank to produce an ordinary Top-1/Top-5 ranking.

Their scaled cosine score is $S_{ij} = \exp(\tau) z_i^\top y_j$, where τ is learnable. Standard independent retrieval ranks every row of S separately. For ground-truth image index t_i, Top-k is

$$\text{Top-}k = \frac{1}{Q} \sum_{i=1}^{Q} \mathbb{1}[t_i \in \text{TopK}_k(S_{i,:})] \,. \tag{2}$$

No knowledge that gallery images are unique is used in Equation 2.

3.2 EEG encoder

The input uses 17 posterior channels and 250 time samples. We flatten the 17×250 response, project it to 512 dimensions, and apply one residual two-layer MLP block with SiLU activation and layer normalization. This intentionally compact architecture limits capacity relative to the image backbone. The output is L2-normalized as in Equation 1. Because the strict A/B/C experiment in Section 5.2 uses this identical EEG architecture in all conditions, its differences cannot be attributed to replacing the EEG encoder. Exact layer shapes, bias and dropout choices, initialization, and trainable-parameter counts are provided in Appendix A.2.

3.3 Conservative image-side LoRA adaptation

The image branch starts from a pretrained CLIP ViT-B/32 image checkpoint. Original CLIP weights remain frozen, and LoRA adapters are inserted into linear modules. The reference setting uses rank $r = 32$, scaling $\alpha = 32$, zero dropout, and Gaussian adapter initialization. The exact checkpoint, image transform, target-module list, initialization semantics, and parameter count are specified in Appendix A.3. If W_0 is a frozen image-layer matrix, the adapted layer is

$$h' = W_0 h + \frac{\alpha}{r} BAh, \tag{3}$$

where A and B are trainable low-rank factors. Unlike frozen-target alignment, Equation 3 allows the image space to move toward EEG-relevant structure while retaining the pretrained backbone.

3.4 Contrastive objective and asymmetric optimization

For a minibatch of N paired EEG–image examples, we minimize symmetric cross-entropy over the similarity matrix:

$$\mathcal{L} = \frac{1}{2N} \sum_{i=1}^{N} \left[-\log \frac{e^{S_{ii}}}{\sum_j e^{S_{ij}}} - \log \frac{e^{S_{ii}}}{\sum_j e^{S_{ji}}} \right]. \tag{4}$$

The reference EEG learning rate is 5×10^{-4}, whereas the image-adapter learning rate is 5×10^{-5}, a ratio of 10:1. Both branches receive gradients from Equation 4, but their update magnitudes differ. This is the operational meaning of asynchronous or two-time-scale learning in our method; it is not alternating training and does not use a separate objective.

3.5 Optional batch-level matching

Given a complete score matrix, independent retrieval uses $\hat{t}_i = \arg\max_j S_{ij}$. Greedy one-to-one decoding processes high-confidence queries first and removes selected images. Hungarian decoding instead solves

$$\hat{\pi} = \arg \max_{\pi \in \mathrm{Perm}(N)} \sum_{i=1}^{N} S_{i,\pi(i)}, \tag{5}$$

for a square session, while rectangular matrices leave surplus rows or columns unmatched. Stable matching uses query-proposing Gale–Shapley preferences derived from S. Sinkhorn applies log-domain scaling to an entropy-regularized transport plan, followed by row-wise plan decoding. We use temperature 0.05, at most 500 iterations, and a strict residual tolerance of 10^{-8}.

These structured methods naturally yield one assignment per query rather than a ranked list. We consequently report their *assignment accuracy*, matched/unmatched counts, and convergence diagnostics. Top-5 is reported only for independent retrieval.

4 Experimental Protocol

4.1 Dataset, splits, and metrics

We use THINGS-EEG2 [1], containing EEG responses to natural object images from ten subjects. Following the standard zero-shot split, training and test concepts are disjoint, and evaluation uses 200 unseen test images. We select 17 posterior channels. Training inputs average four trials; standard test inputs average 80 repeated trials. Subject-dependent experiments train one model per subject. Strict LOSO trains on nine subjects and evaluates the tenth, producing ten folds. Repeated-trial averaging improves signal-to-noise ratio and is explicitly distinguished from single-trial decoding. Appendix A.1 gives the complete split statistics, channel list, time window, and preprocessing audit.

The primary subject-dependent result averages seeds $\{42, 43, 44, 45, 46\}$. We report the mean and sample standard deviation of the five seed-level ten-subject macro means. LOSO and the focused formal confirmation use seed 42 and are labeled single-seed. Random Top-1 and Top-5 levels in a 200-image gallery are 0.5% and 2.5%, respectively.

4.2 Optimization and checkpoint rules

The reference model uses AdamW, cosine learning-rate decay, weight decay 0.05, batch size 512, evaluation batch size 100, 25 epochs, bfloat16, gradient checkpointing, and a gradient-norm limit of 1.0 applied after back-propagation. The main historical grid uses the fixed final epoch to avoid selecting on test data. For fairness-suite NICE and ATM-S re-evaluations, one training run is performed and the checkpoint is selected by validation loss. Our method's fairness matrix reuses its fixed final checkpoint. We report these differences because the suite is a controlled implementation comparison, not a claim of perfect reproduction.

4.3 Non-bijective robustness scenarios

Matching assumptions are tested on identical scores under: the standard 200×200 session; removal of 5 or 10 EEG queries; removal of 5 or 10 images; asymmetric removal of both sides; duplication

Table 1: Per-subject 200-way EEG-to-image retrieval accuracy (%). Intra trains and tests within one subject; Inter uses leave-one-subject-out evaluation. Published comparison rows are contextual. Our intra-subject values average five seeds; our inter-subject values are strict seed-42 LOSO (†).

Method	Metric	Sub1	Sub2	Sub3	Sub4	Sub5	Sub6	Sub7	Sub8	Sub9	Sub10	Avg.
Intra-subject: train and test on the same subject												
NICE [4]	Top-1	13.2	13.5	14.5	20.6	10.1	16.5	17.0	22.9	15.4	17.4	16.1
	Top-5	39.5	40.3	42.7	52.7	31.5	44.0	42.1	56.1	41.6	45.8	43.6
ATM-S [5]	Top-1	25.6	22.0	25.0	31.4	12.9	21.3	30.5	38.8	34.4	29.1	27.1
	Top-5	60.4	54.5	62.4	60.9	43.0	51.1	61.5	72.0	51.5	63.5	58.1
UBP [6]	Top-1	41.2	51.2	51.2	51.1	42.2	57.5	49.0	58.6	45.1	61.5	50.9
	Top-5	70.5	80.9	82.0	76.9	72.8	83.5	79.9	85.8	76.2	88.2	79.7
NeuroBridge [7]	Top-1	50.0	63.2	61.6	61.4	54.8	69.7	62.7	71.2	64.0	73.6	63.2
	Top-5	77.6	90.6	91.1	90.0	85.0	92.9	88.8	95.1	91.0	97.1	89.9
Shallow [8]	Top-1	75.4	87.3	82.9	79.1	74.9	90.2	79.0	86.9	81.3	89.3	82.6
	Top-5	94.3	99.0	**98.3**	**96.5**	96.4	99.2	97.3	99.4	**97.8**	99.2	97.7
Our method	**Top-1**	**84.0**	**89.7**	**85.2**	**83.4**	**84.0**	**92.8**	**84.8**	**90.9**	**80.3**	**91.5**	**86.66**
	Top-5	**96.6**	**99.7**	97.4	**96.5**	**98.3**	**99.8**	**98.0**	**99.9**	97.7	**99.9**	**98.38**
Inter-subject: leave one subject out for test												
NICE [4]	Top-1	7.6	5.9	6.0	6.3	4.4	5.6	5.6	6.3	5.7	8.4	6.2
	Top-5	22.8	20.5	22.3	20.7	18.3	22.2	19.7	22.0	17.6	28.3	21.4
ATM-S [5]	Top-1	10.5	7.1	11.9	14.7	7.0	11.1	16.1	15.0	4.9	20.5	11.9
	Top-5	26.8	24.8	33.8	39.4	23.9	35.8	43.5	40.3	22.7	46.5	33.8
UBP [6]	Top-1	11.5	15.5	9.8	13.0	8.8	11.7	10.2	12.2	15.5	16.0	12.4
	Top-5	29.7	40.0	27.0	32.3	33.8	31.0	23.8	32.2	40.5	43.5	33.4
NeuroBridge [7]	Top-1	23.2	21.2	**13.2**	17.0	14.5	**25.0**	15.3	**20.1**	13.7	27.2	19.0
	Top-5	52.4	49.3	**36.5**	45.3	37.7	**55.0**	45.1	44.9	36.5	56.3	45.9
Shallow [8]	Top-1	24.6	31.3	11.4	19.9	**19.0**	24.1	**18.6**	17.6	**23.3**	**34.6**	**22.4**
	Top-5	54.7	61.5	31.1	48.8	**45.5**	49.8	**51.6**	**46.7**	**54.9**	63.2	**50.8**
Our method†	**Top-1**	**27.0**	**31.5**	13.0	**20.5**	14.0	15.5	17.0	15.0	20.0	30.5	20.40
	Top-5	**59.0**	**63.5**	34.0	**50.5**	37.5	47.5	43.5	42.5	50.5	**64.5**	49.30

of 10 or 20 gallery images; addition of unrelated distractor images; addition of EEG queries whose correct image is absent; and combinations of missing, duplicated, and distractor items.

For a realistic duplicate-EEG condition, we do not copy score rows. Within every recording session, the first ten trials for an image are averaged into EEG-A-1 and the remaining ten into EEG-A-2. The two averages are disjoint real measurements. The base uses 200 EEG-A-1 queries; adding EEG-A-2 for 10 or 20 selected images produces 210×200 and 220×200 matrices. Their absolute scores are compared only within this 20-trial-average suite, not against the standard 80-trial test.

5 Results

5.1 Subject-dependent and cross-subject retrieval

Table 1 follows the requested per-subject format. Literature rows use reported values under closely related 200-way THINGS-EEG2 protocols; encoder, preprocessing, checkpoint, and seed details still differ, so the table is contextual rather than a fully controlled ranking. Our subject-dependent row is the five-seed mean. Our LOSO row is a strict seed-42 re-evaluation and is marked accordingly.

The five seed-level subject-dependent macro means range from 85.60% to 87.35% Top-1 and from 98.20% to 98.55% Top-5, yielding $86.66 \pm 0.69\%$ and $98.38 \pm 0.14\%$. Our method is numerically strongest among the displayed intra-subject rows, but protocol differences preclude an unqualified state-of-the-art claim. In strict LOSO, the macro mean drops to $20.40 \pm 6.91\%$ Top-1 and $49.30 \pm 10.47\%$ Top-5 across held-out subjects. Figure 2 shows that all folds remain above chance but vary strongly, confirming that subject shift is still a central limitation.

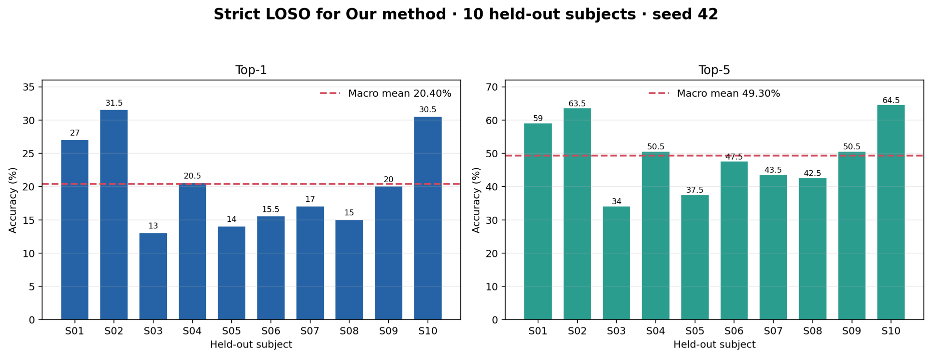

Figure 2: Strict leave-one-subject-out results at seed 42. Bars show independent-retrieval Top-1 and Top-5 for each held-out subject; dashed lines mark macro means. This is a single-seed generalization diagnostic rather than a five-seed population estimate.

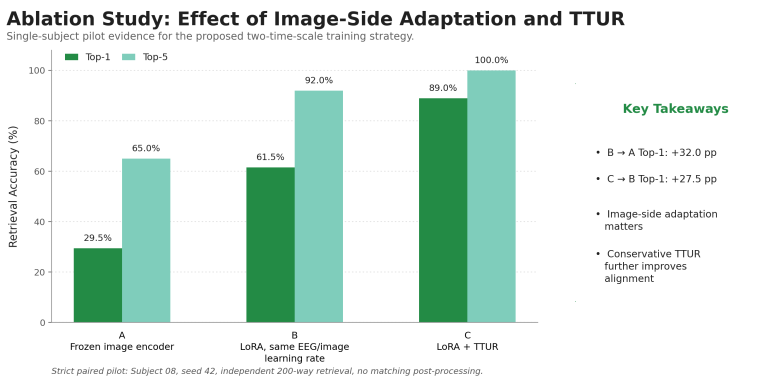

Figure 3: Strict paired subject-08/seed-42 TTUR ablation. All arms use the identical EEG encoder: A freezes the image encoder, B adds equal-rate image-side LoRA, and C uses slower image-side LoRA updates. Values are independent-retrieval accuracy (%).

5.2 Strict paired test of asymmetric co-adaptation

The A/B/C experiment directly addresses whether the improvement is merely caused by our EEG encoder. All three settings use subject 08, seed 42, the same EEG encoder architecture and initialization, identical samples, objective, batch order, and training budget. Only image-side optimization differs. Figure 3 shows that training the EEG encoder alone is insufficient: allowing image-side LoRA adaptation raises Top-1 by 32 points, and slowing the image update relative to the EEG update adds a further 27.5 points.

Relative to A, C gains $+59.5$ Top-1 and $+35.0$ Top-5 points. Because the EEG architecture is controlled, the result supports our proposed innovation: high performance depends on dual-side co-adaptation and its asymmetric update schedule, not only on a newly chosen EEG encoder. It does not, by itself, prove that 10:1 is universally optimal; the next experiment examines nearby settings.

5.3 Rank, learning-rate, and ratio study

The focused validation grid crosses LoRA rank $r \in \{32, 64\}$, EEG learning rate $\eta_e \in \{5 \times 10^{-4}, 10^{-3}\}$, and EEG-to-image learning-rate ratio $\rho \in \{5, 10, 20\}$, for 12 configurations on subject 08/seed 42. Table 2 reports validation scores. The highest validation Top-1 occurs at $(r, \eta_e, \rho) = (64, 10^{-3}, 20)$.

We then compare the selected configuration with the reference $(32, 5 \times 10^{-4}, 10)$ across three subject-08 validation seeds and finally all ten subjects at seed 42. The selected validation mean

		Ratio $= 5$		Ratio $= 10$		Ratio $= 20$	
Rank	EEG LR	Top-1	Top-5	Top-1	Top-5	Top-1	Top-5
32	5×10^{-4}	31.03	59.94	31.52	60.24	29.52	57.64
32	10^{-3}	29.76	57.58	33.76	62.91	32.42	61.82
64	5×10^{-4}	31.27	59.03	32.06	60.91	30.42	59.52
64	10^{-3}	27.39	55.88	34.18	**63.33**	**34.48**	63.03

Table 2: Focused subject-08/seed-42 validation grid (%). Image LR is η_e/ρ. Bold marks the best value in each metric; Top-1 determines the selected setting.

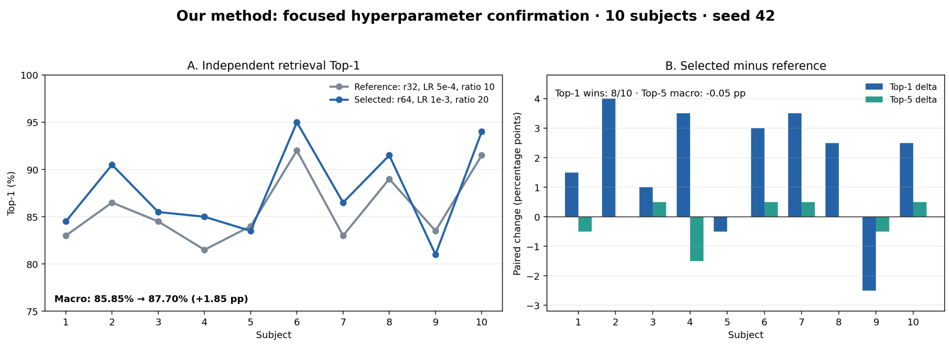

Figure 4: Formal ten-subject seed-42 confirmation of the validation-selected hyperparameter setting against the reference configuration. Top-1 improves by 1.85 points on average, while Top-5 is saturated and essentially unchanged.

is $34.44\%/63.74\%$ versus $31.23\%/60.97\%$ for the reference. In the formal paired test, Figure 4 shows an average Top-1 increase from 85.85% to 87.70%; Top-5 changes from 98.35% to 98.30%. The selected setting wins Top-1 on eight of ten subjects, but it changes rank, absolute rate, and ratio jointly and uses one formal seed. The reference was protocol-locked before this focused study and is the only setting evaluated over all ten subjects and five seeds. We therefore do not retroactively replace the stronger-evidence headline with a post-hoc single-seed follow-up; the grid shows that asymmetric co-adaptation is robust to nearby choices, not that the original 10:1 ratio is globally optimal.

5.4 Fair comparison of matching algorithms

We evaluate NICE, ATM-S, and our method on subject 08/seed 42. Within each baseline, all five decoders receive the exact same EEG queries, image gallery, labels, checkpoint, and score matrix. Thus only the matching rule changes. Table 3 reports the standard 200×200 case. Independent Top-1/Top-5 remains the ordinary retrieval metric; the other columns are assignment accuracy.

The standard session rewards hard global consistency: Hungarian increases assignment accuracy by 13.5, 16.5, and 9.0 points over independent Top-1 for NICE, ATM-S, and our method. This does not mean the representation improved. It means the decoder used the external fact that every query and image should appear exactly once. Applying the same decoder to every baseline makes the comparison fair, but the result remains an offline structured-assignment result.

The non-bijective suite in Figure 5 changes that conclusion. Removing only images creates unanswerable EEG queries; removing only EEG creates distractor gallery images; duplicate gallery entries create equivalent candidates; and combinations alter both matrix shape and answerability. The realistic disjoint-trial experiment is especially revealing. On the 200×200 20-trial-average base, independent/Hungarian/Sinkhorn achieve $90.0/100.0/99.0\%$. At 210×200, they obtain $90.0/92.86/98.10\%$, with Hungarian leaving 10 queries unmatched. At 220×200, they obtain $90.45/89.09/96.36\%$, with Hungarian leaving 20 queries unmatched. The hard bijection is there-

Table 3: Standard 200×200 fairness suite on subject 08/seed 42 (%). Structured columns are assignment accuracy, not retrieval Top-1.

Representation	Indep. T1	Indep. T5	Greedy	Hungarian	Stable	Sinkhorn[‡]
NICE	18.0	41.0	22.0	31.5	23.5	31.5
ATM-S	46.0	76.0	52.0	62.5	50.5	65.0
Our method	**91.0**	**99.5**	**95.5**	**100.0**	**98.0**	**100.0**

[‡]Raw-score Sinkhorn did not meet the preregistered 10^{-8} tolerance within 500 iterations in these three cells; values are diagnostic.

Table 4: Sinkhorn audit over 90 baseline–scenario cells per setting.

Score treatment	Conv. 10^{-6}	Conv. 10^{-8}	Mean assignment
Raw recorded score	30/90	25/90	**62.83%**
Recorded scale removed	90/90	85/90	61.44%
Scale removed + epsilon schedule	90/90	85/90	61.44%

fore beneficial only when its premise is true; multiple legitimate EEG queries per image turn it into a constraint violation.

5.5 Sinkhorn convergence audit

Because Sinkhorn accuracy is meaningful only when its numerical behavior is known, we audit three score treatments over 90 cells each (three baselines by 30 scenarios): raw recorded scores; recorded-scale removal; and scale removal plus an epsilon schedule. Table 4 shows that scale removal greatly improves convergence but does not improve average assignment accuracy.

On the standard matrices, scale removal changes Sinkhorn assignment accuracy from 31.5% to 33.0% for NICE, from 65.0% to 57.0% for ATM-S, and leaves our method at 100.0%. The result rejects the simple hypothesis that better numerical convergence necessarily yields better matching. It also argues against spending the course-project budget on a learned matrix-transform network without a validation-defined objective and leakage-safe training split: convergence is a numerical property, whereas correct assignment depends on calibrated semantic structure.

6 Discussion and Human-Centered Implications

What the system does and does not infer. Our evaluation decodes a stimulus from a known, closed 200-image gallery after repeated presentations. It is not open-world mind reading, does not reconstruct arbitrary thoughts, and does not establish semantic access beyond the experiment. Trial averaging uses substantial repeated evidence, so results should not be presented as single-glance decoding.

Privacy, consent, and agency. EEG is sensitive biometric and health-adjacent data. Deployment would require informed, revocable consent, purpose limitation, secure processing, restricted raw-signal retention, and clear uncertainty communication. Users should control calibration, deletion, and storage. This research system is not a medical device and should not support diagnosis, surveillance, employment screening, or covert inference.

Generalization and evaluation validity. The strict LOSO gap shows that cross-person transfer remains much weaker than subject-dependent calibration. Five seeds quantify optimization variability over the same stimuli, not new participants; literature rows also differ in encoders, preprocessing, and checkpoint rules. The subject-08 matching suite is diagnostic rather than population-level evidence, and batch matching can inflate apparent performance when a gallery is known to be a permutation. We therefore separate retrieval from assignment and test broken assumptions.

Technical limitations. Flattening discards explicit channel topology, and LoRA placement is not neurophysiologically targeted. The focused hyperparameter study is not a full factorial multi-seed search, while Sinkhorn row-wise decoding is a soft many-to-one diagnostic rather than a guaranteed discrete bijection. Future work should prioritize subject-invariant calibration, single-trial robustness, uncertainty, leakage-safe matcher calibration, and prospective studies of usefulness and trust.

7 Conclusion

Our asymmetric EEG–image co-adaptation reaches 86.66% Top-1 and 98.38% Top-5 over ten subjects and five seeds. With the EEG encoder fixed, the strict A/B/C ablation rises from 29.5% to 61.5% and 89.0%, isolating equal-rate and asymmetric image adaptation; focused tuning reaches 87.70%, while strict LOSO falls to 20.40%. Hungarian and Sinkhorn improve valid one-to-one assignments, not embeddings or standard Top-5, and weaken when correspondence fails.

Author contribution statement

- **Chi Kit Wong:** Overall brain–image co-adaptive framework and synchronized asymmetric joint optimization.
- **Sida Zhang:** Session-aware offline decoding as weighted bipartite matching; Hungarian/Sinkhorn analysis.
- **Zhiyi Chen:** Lightweight residual-MLP EEG representation module; hyperparameter analysis.
- **Ziyi Guo:** CLIP/LoRA image adaptation; conservative image-side learning rate.

Equal contribution. All members contributed equally to report writing, result interpretation, revision, and presentation preparation.

Generative-AI disclosure. OpenAI Codex assisted with code/log inspection, LaTeX/Markdown formatting, and language editing, including prompts to audit the page limit, reconcile saved CSV/JSON results, and repair equations. All outputs were reviewed and revised by the authors; drafts are retained in the project task history, and final claims remain the authors' responsibility.

References

[1] Alessandro T. Gifford, Kshitij Dwivedi, Gemma Roig, and Radoslaw M. Cichy. A large and rich EEG dataset for modeling human visual object recognition. *NeuroImage*, 264:119754, 2022. doi: 10.1016/j.neuroimage.2022.119754.

[2] Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations*, 2022.

[3] Martin Heusel, Hubert Ramsauer, Thomas Unterthiner, Bernhard Nessler, and Sepp Hochreiter. GANs trained by a two time-scale update rule converge to a local nash equilibrium. In *Advances in Neural Information Processing Systems*, volume 30, pages 6626–6637, 2017.

[4] Yonghao Song, Bingchuan Liu, Xiang Li, Nanlin Shi, Yijun Wang, and Xiaorong Gao. Decoding natural images from EEG for object recognition. In *International Conference on Learning Representations*, 2024.

[5] Dongyang Li, Chen Wei, Shiying Li, Jiachen Zou, and Quanying Liu. Visual decoding and reconstruction via EEG embeddings with guided diffusion. In *Advances in Neural Information Processing Systems*, volume 37, pages 102822–102864, 2024.

[6] Haitao Wu, Qing Li, Changqing Zhang, Zhen He, and Xiaomin Ying. Bridging the vision–brain gap with an uncertainty-aware blur prior. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 2246–2257, 2025.

[7] Wen Zhang, Shuai Wang, Yu Su, Xin Li, Changqing Zhang, and Shichao Zhong. NeuroBridge: Bio-inspired self-supervised EEG-to-image decoding via cognitive priors and bidirectional semantic alignment. *Proceedings of the AAAI Conference on Artificial Intelligence*, 40(21): 18028–18036, 2026. doi: 10.1609/aaai.v40i21.38863.

[8] Yang Du, Siyuan Dai, Yonghao Song, Paul M. Thompson, Haoteng Tang, and Liang Zhan. Deep models, shallow alignment: Uncovering the granularity mismatch in neural decoding. *arXiv preprint arXiv:2601.21948*, 2026.

[9] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever. Learning transferable visual models from natural language supervision. In *Proceedings of the 38th International Conference on Machine Learning*, volume 139 of *Proceedings of Machine Learning Research*, pages 8748–8763, 2021.

[10] Harold W. Kuhn. The hungarian method for the assignment problem. *Naval Research Logistics Quarterly*, 2(1–2):83–97, 1955.

[11] David Gale and Lloyd S. Shapley. College admissions and the stability of marriage. *The American Mathematical Monthly*, 69(1):9–15, 1962.

[12] Richard Sinkhorn and Paul Knopp. Concerning nonnegative matrices and doubly stochastic matrices. *Pacific Journal of Mathematics*, 21(2):343–348, 1967.

[13] Marco Cuturi. Sinkhorn distances: Lightspeed computation of optimal transport. In *Advances in Neural Information Processing Systems*, volume 26, 2013.

A Additional Experimental Details

This appendix records the implementation details needed to interpret and reproduce the reported experiments. In particular, it distinguishes the upstream 250-Hz whitened EEG derivative from the transformations performed by our own data loader.

A.1 THINGS-EEG2 dataset and EEG preprocessing

THINGS-EEG2 contains ten participants and uses disjoint training and test concepts [1]. Table 5 summarizes the split used in this work. Counts are per participant; consequently, the training and test partitions contain 66,160 and 16,000 recorded trials per participant before averaging.

Table 5: THINGS-EEG2 split statistics used by our retrieval experiments. Training and test concepts do not overlap. Counts are per participant.

Property	Training split	Test split
Participants	10	10
Object concepts	1,654	200
Images per concept	10	1
Unique stimulus images	16,540	200
EEG repetitions per image	4	80
EEG trials per participant	66,160	16,000

Across both splits, this yields 16,740 unique stimulus images and 82,160 EEG trials per participant.

Operational tensors and channels. For each participant, the serialized derivative tensors have shapes $16{,}540 \times 4 \times 63 \times 250$ for training and $200 \times 80 \times 63 \times 250$ for test (image, repetition, channel, time). The model retains the following 17 posterior electrodes, in this exact order: P7, P5, P3, P1, Pz, P2, P4, P6, P8, PO7, PO3, POz, PO4, PO8, O1, Oz, and O2. The original acquisition rate was 1,000 Hz. The local derivative is resampled to 250 Hz. Under the accompanying 250-Hz derivative convention, the model's $[0, 250)$ slice represents the post-stimulus interval $[0, 1{,}000)$ ms. Table 6 separates upstream preparation from the operations executed inside our project.

Table 6: EEG preprocessing audit. "Upstream" denotes preparation of the local `Preprocessed_data_250Hz_whiten` derivative; "project-side" denotes operations executed by our training and evaluation code.

Step	Applied operation
Filtering and sampling	EEG was acquired at 1,000 Hz with the dataset's online 0.1–100-Hz filter [1]. The upstream derivative resamples to 250 Hz. Our loader applies no additional temporal filtering or resampling.
Baseline and crop	The accompanying derivative implementation epochs -200 to $1{,}000$ ms, subtracts the -200 to 0-ms mean separately for each epoch and channel, and then discards the prestimulus samples. Our model receives 250 post-stimulus samples and applies no further baseline correction.
Artifact handling	The dataset paper reports no further artifact correction after target-event exclusion [1]; neither the accompanying derivative implementation nor the project loader adds ICA or amplitude-threshold rejection.
Normalization	The accompanying derivative implementation applies session-wise multivariate noise normalization (MVNN) using a shrinkage-covariance transform across channels, with covariance estimated from the training partition only. This is not independent per-channel z-scoring. The project loader adds no channel-wise standardization.
Trial averaging	The loader takes an unweighted arithmetic mean along the repetition axis. Thus, each training item averages its four stored trials, and each standard test query averages its 80 stored trials. No session-dependent weighting or post-average renormalization is used.
Data augmentation	No explicit EEG augmentation and no stochastic image augmentation are used. Dataset shuffling and minibatch sampling are not treated as augmentation.

The tensor dimensions and the 250-Hz/whitened designation are corroborated by the repository's validated runtime records and directory contract; the `.pt` files themselves do not contain a command-level preprocessing manifest. Baseline correction and MVNN details are therefore attributed to the accompanying derivative-preprocessing implementation rather than claimed as byte-level provenance. Channel selection, temporal slicing, trial averaging, and the absence of additional normalization or stochastic augmentation were verified from the project loader and training/evaluation entry points. For comparison, the standard BioTrain/BioTest matrices use a -200 to 800-ms epoch, 100-Hz sampling, 17 posterior channels, and session-wise MVNN [1]; our experiments instead consume the separate 0–1,000-ms, 250-Hz, 63-channel derivative before project-side channel selection.

A.2 EEG encoder specification

Table 7 expands the phrase "residual two-layer MLP" used in the main text. The input projection precedes one identity-skip residual block; it is not part of the two-layer feed-forward branch.

Table 7: Exact EEG encoder used by the reported formal runs. Parameter counts include trainable weights and biases.

Stage	Input $\to$ output	Bias/affine	Operation	Parameters
Reshape	$17 \times 250 \to 4{,}250$	–	Flatten	0
Input projection	$4{,}250 \to 512$	Yes	Linear	2,176,512
Residual FFN–1	$512 \to 512$	No	Linear, SiLU	262,144
Regularization	$512 \to 512$	–	Dropout $p = 0.1$	0
Residual FFN–2	$512 \to 512$	No	Linear	262,144
Skip and norm	$512 \to 512$	LN only*	Add, LayerNorm	1,024
EEG encoder total				**2,701,824**

*LayerNorm has trainable scale and offset; the identity skip has no parameters. LayerNorm uses $\epsilon = 10^{-6}$.

There is no activation immediately after the input projection, no learned skip projection, and no BatchNorm. Dropout appears only between SiLU and the second residual-branch linear layer. The 512-dimensional encoder output is L2 normalized outside the encoder before similarity computation. Model initialization follows the pinned Transformers `post_init` path: linear weights are sampled from $\mathcal{N}(0, 0.02^2)$, linear biases are zero, and LayerNorm scale and offset are initialized to one and zero. The joint brain-side module additionally learns one scalar logit scale initialized to 2.0, giving 2,701,825 trainable brain-side parameters when that scalar is included.

A.3 Image encoder and LoRA specification

The exact base checkpoint is `laion/CLIP-ViT-B-32-laion2B-s34B-b79K`, revision `1a25a446712ba5ee05982a381eed697ef9b435cf`. These are LAION-2B-trained OpenCLIP weights consumed through Hugging Face Transformers' `CLIPVisionModelWithProjection`; they are not the original OpenAI ViT-B/32 weights. The vision transformer uses 224-pixel inputs, 32×32 patches, 12 transformer blocks, hidden width 768, MLP width 3,072, 12 attention heads, and a 512-dimensional output projection.

The associated `CLIPImageProcessor` converts input images to RGB, resizes the shortest edge to 224 with bicubic interpolation, center-crops to 224×224, rescales pixels by $1/255$, and normalizes RGB channels with mean $(0.48145466, 0.45782750, 0.40821073)$ and standard deviation $(0.26862954, 0.26130258, 0.27577711)$. No random crop, flip, or color jitter is applied.

The reference model applies rank-32 LoRA to every `Linear` module in the image encoder. This comprises Q, K, V, and output attention projections and both MLP projections in each transformer block, plus the final visual projection. Patch convolution and LayerNorm modules are excluded. Table 8 gives the resulting module and parameter counts.

For every adapted linear layer, $r = 32$, $\alpha = 32$, LoRA dropout is zero, and no LoRA bias is trained, so the multiplier α/r equals one. PEFT's `gaussian` rule initializes A from a zero-mean Gaussian with standard deviation $1/r = 0.03125$ and initializes B to zero. Consequently, the initial LoRA residual BA is exactly zero, so the wrapped model initially matches the frozen base under the same

Table 8: Rank-32 LoRA placement and trainable image-side parameter count. A linear layer $d_{\text{in}} \to d_{\text{out}}$ contributes $r(d_{\text{in}} + d_{\text{out}})$ adapter parameters.

Target group	Modules	Dimensions	LoRA parameters
Q/K/V/output projections	48	$768 \to 768$	2,359,296
MLP `fc1`/`fc2`	24	$768 \leftrightarrow 3{,}072$	2,949,120
Final visual projection	1	$768 \to 512$	40,960
Total	73	–	**5,349,376**

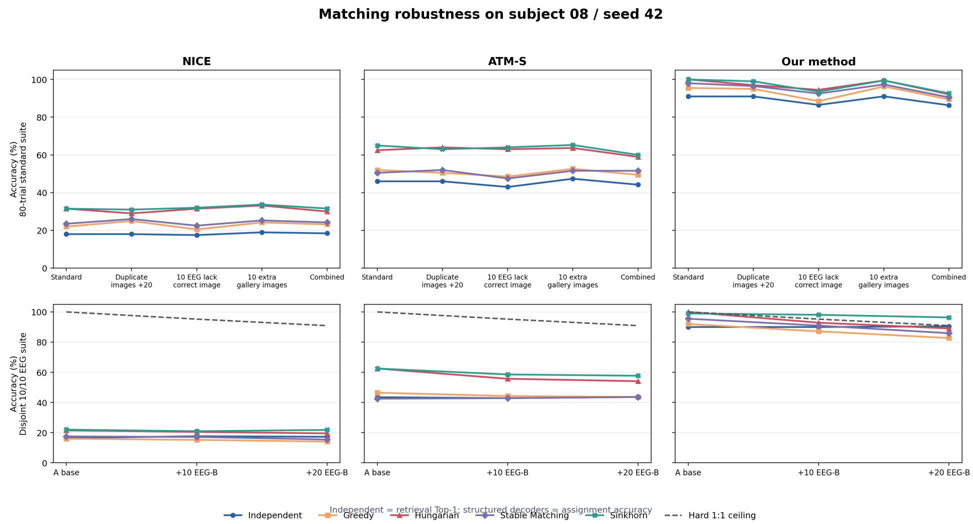

Figure 5: Matching robustness under missing, duplicate, and distractor items, including disjoint-trial duplicate EEG. Hard one-to-one assignment exposes unmatched queries when bijection is false.

numerical precision and preprocessing. Combining the 5,349,376 image-side adapter parameters with the 2,701,825 brain-side parameters (including logit scale) gives **8,051,201** trainable parameters in the reference joint model; all original image-encoder parameters remain frozen.

A.4 Additional Results